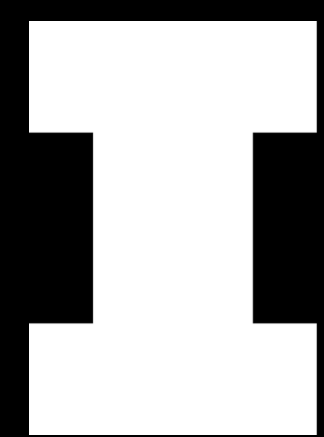


Invasion, hybridization, and speciation: The breakdown of reproductive isolating barriers in the San Francisco Bay

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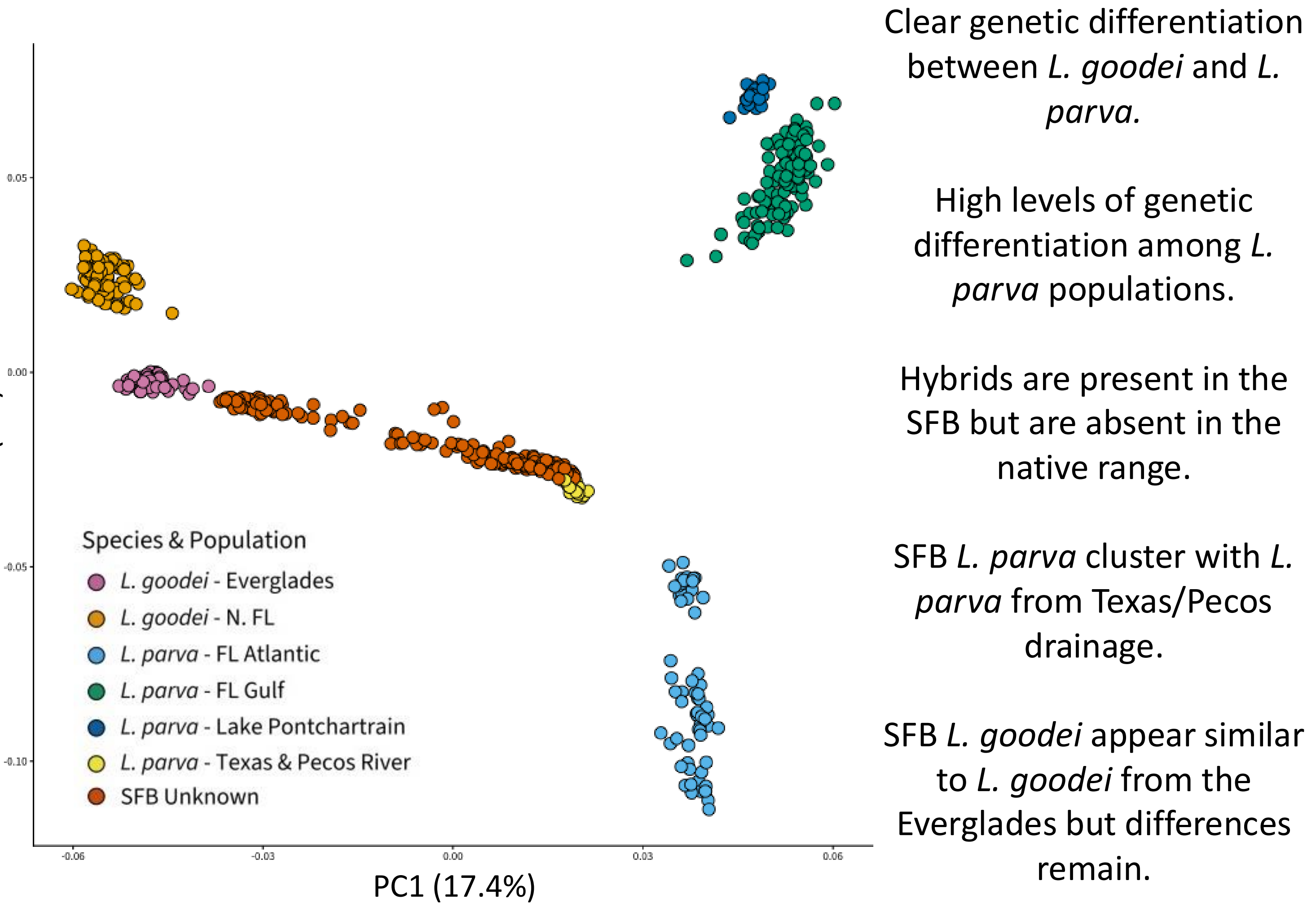
Introduction

- Hybridization between previously isolated lineages, once thought rare, is now recognized as a regular consequence of anthropogenic disturbance¹.
- Species introduction is a chief driver, rapidly eroding reproductive isolating barriers (RIBs) that took millions of years to evolve.
- Understanding which features of novel environments and invaders facilitate RIB breakdown is both a conservation concern and a window into how RIBs evolve, are maintained, and can be disrupted.
- Lucania goodei* and *L. parva* are sister species with high behavioral isolation in native areas of sympatry, due to reinforcement².
- Both species have been introduced to the San Francisco Bay (SFB): *L. parva* in the 1950s (now widely abundant) and *L. goodei* in 2017. Intermediate-appearing fish have recently been observed³.
- We ask: Has reproductive isolation broken down in the SFB, and if so, did the source fish originate from allopatric or sympatric populations?**

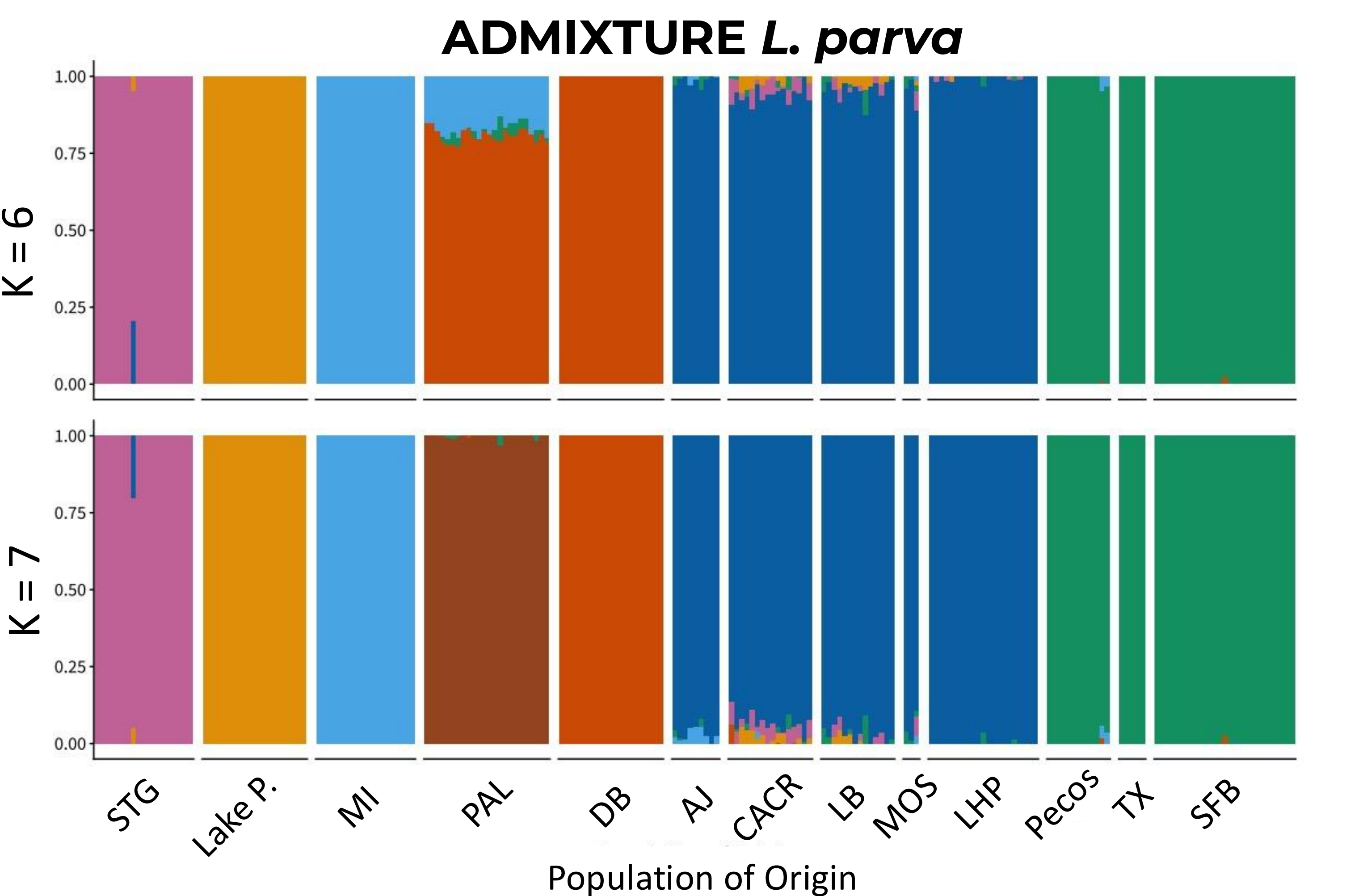
Methods

- Fish were collected from the SFB in 2024; Fish from FL, LA, and TX were collected in 2017.
- All were sequenced with RAD-Seq.
- Loci were called with STACKS. The STACKS populations module was run multiple times, followed by further filtering with vcftools and plink.
- PCA and ADMIXTURE were used to identify hybrids and determine populations of origin.

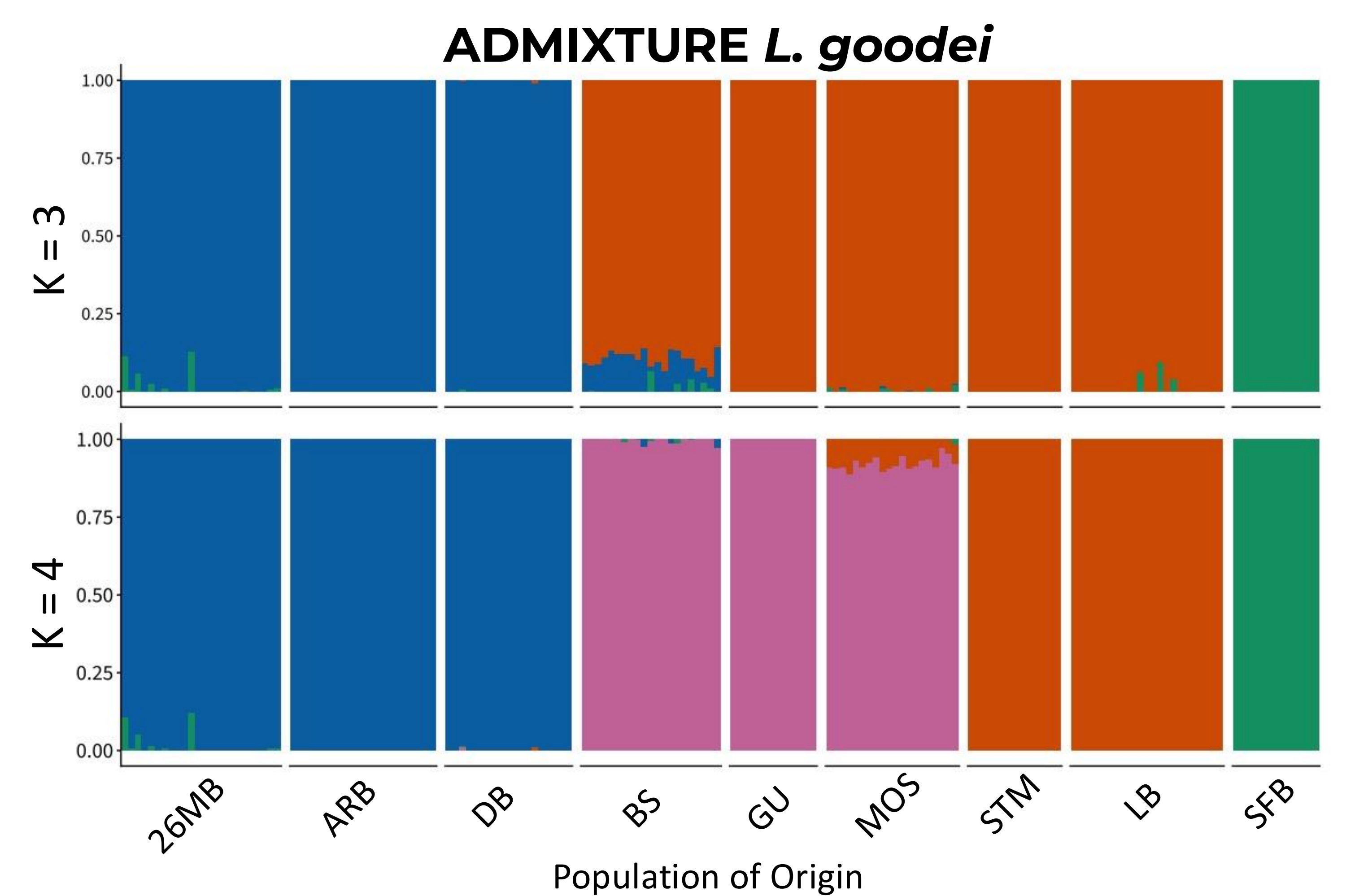
Results



Results

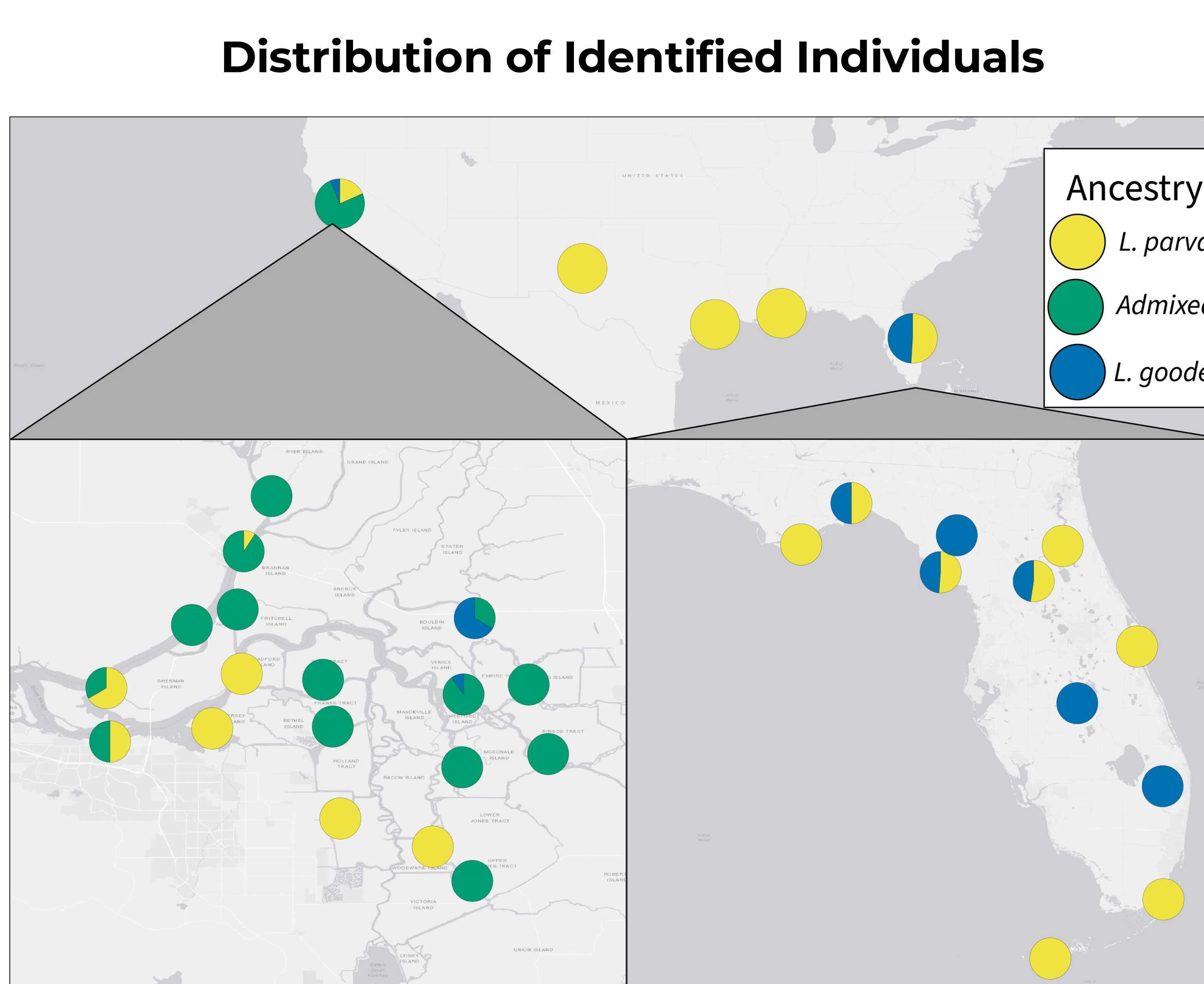


A K (the number of hypothesized ancestral populations) of 6 was best supported for *L. parva*, the theoretical cluster shared by the Pecos River, Texas, and SFB fish was maintained even in the second-best supported K of 7.



A K of 4 was best supported for *L. goodei*, even when lowering K to 3, the SFB cluster remained sufficiently distinct from any of our sequenced populations.

Results



Discussion

- Contemporary hybridization is prevalent and ongoing in the San Francisco Bay between *L. goodei* and *L. parva*. There is little evidence for hybridization in the native range.
- The source population for *L. parva* is likely from the Pecos River Drainage, which is allopatric to *L. goodei*. The source for *L. goodei* is unclear but likely from South-West Florida.
- Previous work shows that behavioral isolation is lowest when allopatric *L. parva* males interact with allopatric *L. goodei* females^{2,4}. These conditions are at least, partially met in this system.
- The presence of admixed individuals throughout the study area suggests that the ecological conditions that disfavor hybrids in the native range are absent in the San Francisco Bay.

Acknowledgements - Citations

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1. Rhymer, J. M. & Simberloff (1996) *Annu. Rev. Ecol. Syst.* 27 2. Fuller, R. C., McGhee, K. E. & Schrader (2007) *J. Evol. Biol.* 20 3. Mahardja, B. *et al.* (2020) *San Franc. Estuary Watershed Sci.* 18 4. St. John, M. E. & Fuller, R. C. (2021) *Curr. Zool.* 67

